ℂomplex Eigenvalue Field extensions of Fibonacci Matrices

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Abstract

The Fibonacci sequence, $\mathcal{F}_n$, is well-known for its closed-form expression, Binet's formula, which links the sequence to the Golden Ratio, ϕ. This paper explores the algebraic framework of Fibonacci matrices M, focusing on their diagonalization via characteristic polynomials. Extending M into the complex field $\mathbb{M}$ enables the study of eigenvalues, eigenvectors, and Jordan decomposition. The recursive nature of the sequence is shown to arise from the determinant operation on $n \times n$ complex tridiagonal matrices, whose eigenvalues are $1 + 2i\cos\big(\pi k/(n+1)\big)$ and so lie in algebraic field extensions of $\mathbb{Q}$ governed by the integer $n+1$. When $d+1$ divides $n+1$ the eigenvalues of the smaller matrix reappear in the larger, the matrix form of the divisibility of $\mathcal{F}_{n+1}$ by $\mathcal{F}_{d+1}$, and when $n+1$ is prime the eigenvalues require the least familiar radicals. This primarily pedagogical piece introduces standard algebraic operations in novel contexts and provides an entry point for further study of field extensions and polynomial properties.

The **Fibonacci sequence**, $\mathcal{F}_n$ possesses a closed-form expression known as **Binet's formula**, which represents each Fibonacci number as a function of the Golden Ratio $\phi = \frac{1+\sqrt{5}}{2}$, encapsulating the sequence's asymptotic behavior towards ϕ. Fibonacci matrices, M provide an algebraic framework to analyze and compute Fibonacci numbers efficiently. This paper focuses on the diagonalisation of M, through its **characteristic polynomial**. An extension into complex fields, $\mathbb{M}$ affords an exploration of eigenvalues, eigenvectors, and the process of Jordan decomposition.

$\mathcal{F}_n$'s second cousin, the **Pell sequence**, $\mathcal{S}_n$ is useful in various number-theoretic contexts, including the approximation of square roots and the construction of Pell primes, which satisfy certain divisibility properties that make them of interest in primality testing [1]. $\mathcal{S}_n$ also has a closed-form expression similar to Binet's formula,[2].

Both sequences provide opportunities to inform the inquisitive high school thinker of some standard operations applied in novel circumstances. As such this is mostly a pedagogical piece with a few suggestive elements from **algebraic field extension** theory that invite the reader to explore further. Recursive sequences are generated by the action of the determinant operation on $n \times n$ ℂomplex **tridiagonal matrices** [3]. The **Jordan diagonalised** representation of these matrices carries their eigenvalues on its diagonal, and the product of those eigenvalues, the determinant, is the Fibonacci number $\mathcal{F}_{n+1}$. The eigenvalues take the structured complex radical form $1 + 2i\cos\big(\pi k/(n+1)\big)$, and the field needed to write them down is decided by the integer $n+1$: when $n+1$ has many small divisors the eigenvalues of smaller Fibonacci matrices reappear and the radicals stay simple, and when $n+1$ is prime a genuinely new layer of algebraic extension of $\mathbb{Q}$ is required. The Python notebook generalEigenVectorFibonacci.ipynb computes the diagonal forms, and the cases that sympy's `simplify` leaves unsimplified invite a Galois-type investigation of their polynomial, field-extended, properties.

Characteristic Matrices for Metallic Sequences

The Fibonacci sequence $\mathcal{F}_n$ defined by the recurrence relation, $\mathcal{F}_n = \mathcal{F}_{n-1} + \mathcal{F}_{n-2}$, with initial conditions $\mathcal{F}_0 = 0$ and $\mathcal{F}_1 = 1$ can be generated with the repeated application of the golden **Fibonacci matrix** M_1 :

$$M_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \tag{1}$$

on the vector $\begin{pmatrix} \mathcal{F}_{n+1} \\ \mathcal{F}_n \end{pmatrix}$ as:

$$\begin{pmatrix} \mathcal{F}_{n+1} \\ \mathcal{F}_n \end{pmatrix} = M_1^n \begin{pmatrix} \mathcal{F}_1 \\ \mathcal{F}_0 \end{pmatrix}.$$

The **characteristic polynomial** of M_1:

$$\det(M_1 - \lambda 1) = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 1,$$

solves the *characteristic equation*, $\lambda^2 - \lambda - 1 = 0$, with **eigen value** roots:

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \quad \lambda_2 = \frac{1-\sqrt{5}}{2}.$$

The Fibonacci generator matrix (1) can be diagonalized as:

$$M_1 = PDP^{-1},$$

where P is the matrix of eigenvectors, and D is the diagonal matrix of eigenvalues. See appendix for details. Complex Fibonacci matrices introduce imaginary components in the recurrence relations:

$$\mathbb{M} = \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix},$$

with eigenvalues $\lambda = 1 \pm i$ from a characteristic polynomial:

$$\det(\mathbb{M} - \lambda I) = \lambda^2 - 2\lambda + 2.$$

We can generalize the Fibonacci recurrence using a *metallic* multiplier parameter m that adjusts the contribution of the prior term in the sequence. The modified recurrence relation takes the form:

$$\mathcal{S}_n = m \cdot \mathcal{S}_{n-1} + \mathcal{S}_{n-2},$$

or through a matrix M_m that varies depending on the choice of m:

$$M_m = \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}.$$

With $m = 2$ and initial values $\mathcal{S}_0 = 1$ and $\mathcal{S}_1 = 1$, the matrix M_2 produces a sequence of Pell (silver) type:

$$\begin{pmatrix} \mathcal{S}_2 \\ \mathcal{S}_1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} \mathcal{S}_3 \\ \mathcal{S}_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix}.$$

The resulting terms $\mathcal{S}_0 = 1$, $\mathcal{S}_1 = 1$, $\mathcal{S}_2 = 3$, $\mathcal{S}_3 = 7$, $\mathcal{S}_4 = 17$, $\mathcal{S}_5 = 41$ are the numerators of the continued-fraction approximations $1, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}$ to $\sqrt{2}$, sometimes called the half companion Pell numbers; starting instead from $\mathcal{S}_0 = 0$, $\mathcal{S}_1 = 1$ gives the Pell numbers proper, $0, 1, 2, 5, 12, 29, \ldots$, whose terms are the denominators of the same approximations. Both share the characteristic equation arising from the characteristic polynomial of M_2:

$$\det(M_2 - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 2\lambda - 1,$$

(where I is the identity matrix), whose roots are the silver ratio and its conjugate:

$$\lambda_1 = 1 + \sqrt{2}, \quad \lambda_2 = 1 - \sqrt{2}.$$

Complex Matrix Generator of the Fibonacci Sequence

Geometrically the **determinant** of a matrix of row vectors, $\mathbf{a}, \mathbf{b}, \mathbf{c}$ expresses their cross and dot products, which for $3d$ looks like:

$$\det \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \\ \mathbf{c} \end{pmatrix} = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}),$$

where the *cross product* $\mathbf{b} \times \mathbf{c}$ produces a vector orthogonal to both $\mathbf{b}$ and $\mathbf{c}$, and the *dot product* $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ measures how much of $\mathbf{a}$ aligns with this orthogonal vector. Consider first the simplest case of a 2 row, $\mathbb{C}$omplex-valued square matrix $\mathbb{M}(2)_m$:

$$\mathbb{M}(2)_m = \begin{pmatrix} 1 & m \\ m & 1 \end{pmatrix}, \tag{2}$$

whose determinant is $\det \mathbb{M}(2)_m = 1 \times 1 - m \times m = 1 - m^2$. We require this to yield the third Fibonacci number $\mathcal{F}_3 = 2$, so $1 - m^2 = 2 \Rightarrow m = \pm i$, and choosing $m = i$ we have as our 2×2 Fibonacci matrix:

$$\mathbb{M}(2)_1 = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}. \tag{3}$$

The off-diagonal elements i represent a phase shift, analogous to the anti-symmetric interactions in the cross-product. We extend this to a 3×3 matrix $\mathbb{M}(3)_1 \equiv \mathbb{M}_3$ to ensure that the matrix begins to propagate the Fibonacci sequence correctly:

$$\mathbb{M}(3)_1 = \begin{pmatrix} 1 & m & 0 \\ m & 1 & m \\ 0 & m & 1 \end{pmatrix}.$$

Setting the determinant of $\mathbb{M}_3$ to 3 (the fourth Fibonacci number) we have, expanding along the first row,

$$\det \mathbb{M}(3)_1 = 1 \cdot (1 - m^2) - m\,(m \cdot 1 - m \cdot 0) = 1 - 2m^2 = 3, \quad \Rightarrow m = \pm i.$$

The same expansion along the last column, applied to the $n \times n$ matrix with 1 on the diagonal and i beside it, gives $\det \mathbb{M}_n = \det \mathbb{M}_{n-1} - i^2 \det \mathbb{M}_{n-2} = \det \mathbb{M}_{n-1} + \det \mathbb{M}_{n-2}$, so that $\det \mathbb{M}_n = \mathcal{F}_{n+1}$ for every n.

Extension to higher *metallic numbers* m promotes the diagonal from 1 to m, keeping i beside it, so that the metallic matrix $\mathbb{M}(n)_m$ is

$$\mathbb{M}(n)_m = \begin{pmatrix} m & i & 0 & \cdots & 0 \\ i & m & i & \cdots & 0 \\ 0 & i & m & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & i \\ 0 & 0 & 0 & i & m \end{pmatrix}, \qquad \det \mathbb{M}(n)_m = m \det \mathbb{M}(n-1)_m + \det \mathbb{M}(n-2)_m,$$

which is the metallic recurrence $\mathcal{S}_n = m\,\mathcal{S}_{n-1} + \mathcal{S}_{n-2}$. Placing m off the diagonal instead, as mi, would give $\det \mathbb{M}_n = \det \mathbb{M}_{n-1} + m^2 \det \mathbb{M}_{n-2}$, whose characteristic equation $x^2 = x + m^2$ is not the metallic one. For the Pell sequence (silver series, $m = 2$) the $n = 3$ matrix reads:

$$\mathbb{M}(3)_2 = \begin{pmatrix} 2 & i & 0 \\ i & 2 & i \\ 0 & i & 2 \end{pmatrix},$$

whose determinant yields the fourth number of the Pell sequence proper, $2, 5, 12, 29, \ldots$:

$$\det \mathbb{M}(3)_2 = 2(2 \times 2 - i \times i) - i(i \times 2 - i \times 0) = 10 + 2 = 12.$$

Complex Binet & diagonalized Jordan form

To diagonalize the matrix (3), $\mathbb{M}(2)_1 = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$, we start by finding the eigenvalues $\lambda = 1 \pm i \in \mathbb{Q}(i)$ that solve the characteristic polynomial:

$$\det(\mathbb{M}(2)_1 - \lambda I) = \det \begin{pmatrix} 1-\lambda & i \\ i & 1-\lambda \end{pmatrix} = (1-\lambda)^2 - i^2 = \lambda^2 - 2\lambda + 2.$$

To find the eigenvector corresponding to $\lambda_1 = 1 + i$, we solve $(\mathbb{M}(2)_1 - \lambda_1 I)\mathbf{v} = 0$.

$$\mathbb{M}(2)_1 - \lambda_1 I = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} - \begin{pmatrix} 1+i & 0 \\ 0 & 1+i \end{pmatrix} = \begin{pmatrix} -i & i \\ i & -i \end{pmatrix}.$$

Let $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, then:

$$\begin{pmatrix} -i & i \\ i & -i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives the equations:

$$-iv_1 + iv_2 = 0 \quad \Rightarrow \quad v_1 = v_2.$$

Thus, an eigenvector for $\lambda_1 = 1 + i$ is $\mathbf{v}_{\lambda_1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Similarly, $\mathbf{v}_{\lambda_2} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Since $\mathbb{M}(2)_1$ has distinct eigenvalues, we look to express it as $\mathbb{M}(2)_1 = PJP^{-1}$ being diagonalizable[1] as J, by defining P from the column eigen vectors and J_2 from its two eigenvalues:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}.$$

[1]For higher order or complex matrices, Jordan decomposition generalizes the diagonalization, $\mathbb{M} = PJP^{-1}$.

The determinant of J_2 is the product of its diagonal entries:

$$\det J_2 = (1-i)\cdot(1+i) = 1-(i)^2 = 1+1 = 2.$$

We determine P^{-1} given the determinant, $\det(P) = 1\cdot(-1) - 1\cdot 1 = -2$ as

$$P^{-1} = \frac{1}{\det(P)}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Expanding PJ_2P^{-1}:

$$\mathbb{M}(2)_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

Field Splitting within $\det \mathbb{M}_n$

Now starting from the complex matrix generator for Fibonacci (we will drop the subscript 1 in the following as all sequences are Fibonacci-metal, $m = 1$):

$$\mathbb{M}(3)_1 \equiv \mathbb{M}_3 = \begin{pmatrix} 1 & i & 0 \\ i & 1 & i \\ 0 & i & 1 \end{pmatrix},$$

we now seek its Jordan form J_3, as well as the transformation matrix P such that $\mathbb{M}_3 = PJ_3P^{-1}$. The characteristic polynomial for $\mathbb{M}_3$ reads:

$$\lambda^3 - 3\lambda^2 + 5\lambda - 3 = 0$$

Solving this polynomial yields one real and two complex eigenvalues built from $\sqrt{2}$, the irrational part of the **silver ratio**:

$$\lambda_1 = 1, \quad \lambda_2 = 1 - \sqrt{2}\cdot i, \quad \lambda_3 = 1 + \sqrt{2}\cdot i$$

Since each eigenvalue has multiplicity 1, the Jordan form J_3 is a diagonal matrix:

$$J_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1-\sqrt{2}\cdot i & 0 \\ 0 & 0 & 1+\sqrt{2}\cdot i \end{pmatrix}.$$

The determinant of J_3, whose entries lie in $\mathbb{Q}(i, \sqrt{2})$, is the product of its diagonal entries:

$$\det J_3 = 1\cdot\left(1-\sqrt{2}\,i\right)\cdot\left(1+\sqrt{2}\,i\right) = 3.$$

The eigenvectors corresponding to each eigenvalue are:

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix},$$

forming the columns of the matrix P which has an inverse P^{-1}:

$$P = \begin{pmatrix} -1 & 1 & 1 \\ 0 & -\sqrt{2} & \sqrt{2} \\ 1 & 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} -\frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & -\frac{\sqrt{2}}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{\sqrt{2}}{4} & \frac{1}{4} \end{pmatrix}$$

To confirm the Jordan decomposition, we can calculate PJP^{-1} and verify that it reconstructs $\mathbb{M}_3$. You can do this and find subsequent diagonal matrices by using the python code, generalEigenVectorFibonacci.ipynb. In every case

$$\det(J_n) = \mathcal{F}_{n+1}.$$

We can proceed with this analysis gradually building up the order n of the characteristic polynomial, as shown in Table 1, written as $\det(\lambda I - \mathbb{M}_n)$ so that each polynomial is monic; its constant term is $(-1)^n\mathcal{F}_{n+1}$, and setting $\lambda = 0$ recovers the determinant.

n	**Characteristic Polynomial**	$\mathcal{F}_{n+1}$
2	$\lambda^2 - 2\lambda + 2$	2
3	$\lambda^3 - 3\lambda^2 + 5\lambda - 3$	3
4	$\lambda^4 - 4\lambda^3 + 9\lambda^2 - 10\lambda + 5$	5
5	$\lambda^5 - 5\lambda^4 + 14\lambda^3 - 22\lambda^2 + 20\lambda - 8$	8
6	$\lambda^6 - 6\lambda^5 + 20\lambda^4 - 40\lambda^3 + 51\lambda^2 - 38\lambda + 13$	13
7	$\lambda^7 - 7\lambda^6 + 27\lambda^5 - 65\lambda^4 + 105\lambda^3 - 111\lambda^2 + 71\lambda - 21$	21
8	$\lambda^8 - 8\lambda^7 + 35\lambda^6 - 98\lambda^5 + 190\lambda^4 - 256\lambda^3 + 233\lambda^2 - 130\lambda + 34$	34
9	$\lambda^9 - 9\lambda^8 + 44\lambda^7 - 140\lambda^6 + 315\lambda^5 - 511\lambda^4 + 594\lambda^3 - 474\lambda^2 + 235\lambda - 55$	55
10	$\lambda^{10} - 10\lambda^9 + 54\lambda^8 - 192\lambda^7 + 490\lambda^6 - 924\lambda^5 + 1295\lambda^4 - 1324\lambda^3 + 942\lambda^2 - 420\lambda + 89$	89
11	$\lambda^{11} - 11\lambda^{10} + 65\lambda^9 - 255\lambda^8 + 726\lambda^7 - 1554\lambda^6 + 2534\lambda^5$ $-3130\lambda^4 + 2860\lambda^3 - 1836\lambda^2 + 744\lambda - 144$	144

Table 1: Characteristic Polynomials and Fibonacci Numbers

Why the eigenvalues are cosines

The pattern in the eigenvalues becomes transparent once $\mathbb{M}_n$ is written as $I + iP_n$, where P_n is the adjacency matrix of a path of n points, with 1 beside the diagonal and 0 elsewhere. The vector with entries $\sin(xa)$, for $x = 1, \ldots, n$ and $a = \pi k/(n+1)$, is a standing wave on the path, and the identity $\sin(x-1)a + \sin(x+1)a = 2\cos a \sin xa$ shows that P_n multiplies it by $2\cos a$. Every eigenvalue of $\mathbb{M}_n$ is therefore of the form

$$\lambda_k = 1 + 2i\cos\frac{\pi k}{n+1}, \qquad k = 1, \ldots, n,$$

so that all of them lie on the vertical line $\operatorname{Re}\lambda = 1$, between $1 - 2i$ and $1 + 2i$, and pairing k with $n + 1 - k$ pairs each with its complex conjugate. The eigenvectors are the same standing waves, independent of the choice of diagonal; for $n = 3$ they are the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ above. Two consequences organise everything that follows.

First, the numbers needed to write down the eigenvalues are the numbers needed to write down $\cos\big(\pi/(n+1)\big)$, which has degree $\varphi\big(2(n+1)\big)/2$ over $\mathbb{Q}$, where φ is Euler's totient function. The

degree is 2 for $n = 4$, where $2\cos(\pi/5)$ is the golden ratio, but it is 3 for $n = 6$ and $n = 8$ and 5 for $n = 10$, and this, rather than any property of n itself, is what makes the eigenvalues for those orders resist simplification.

Second, if $d+1$ divides $n+1$, every angle $\pi k/(d+1)$ is also one of the angles $\pi k'/(n+1)$, so the eigenvalues of $\mathbb{M}_d$ reappear among those of $\mathbb{M}_n$. The leftover eigenvalues multiply to $\mathcal{F}_{n+1}/\mathcal{F}_{d+1}$, and this is the matrix form of the classical fact that $\mathcal{F}_{d+1}$ divides $\mathcal{F}_{n+1}$ whenever $d+1$ divides $n+1$.

Small cases, organised by $n+1$

The 5×5 matrix, with $n+1 = 6$, inherits 1 from $\mathbb{M}_1$ and $1 \pm i$ from $\mathbb{M}_2$, and adds complex eigenvalues built from $\sqrt{3}$, $\lambda \in \mathbb{Q}(i, \sqrt{3})$:

$$J_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1-i & 0 & 0 & 0 \\ 0 & 0 & 1+i & 0 & 0 \\ 0 & 0 & 0 & 1-\sqrt{3}\,i & 0 \\ 0 & 0 & 0 & 0 & 1+\sqrt{3}\,i \end{pmatrix}$$

with determinant $\det J_5 = (1) \cdot (1-i) \cdot (1+i) \cdot \left(1-\sqrt{3}\,i\right) \cdot \left(1+\sqrt{3}\,i\right) = 8$.
The 7×7 matrix, with $n+1 = 8$, inherits the eigenvalues of J_3, which involve $\sqrt{2}$, since 4 divides 8, and adds $1 \pm i\sqrt{2 \pm \sqrt{2}}$, since $2\cos(\pi/8) = \sqrt{2+\sqrt{2}}$ and $2\cos(3\pi/8) = \sqrt{2-\sqrt{2}}$:

$$J_7 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1-\sqrt{2}\,i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1+\sqrt{2}\,i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-i\sqrt{2+\sqrt{2}} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1+i\sqrt{2+\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-i\sqrt{2-\sqrt{2}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1+i\sqrt{2-\sqrt{2}} \end{pmatrix}$$

For $n = 4$ and $n = 9$ we define the real irrational coefficients

$$\alpha_\pm = \sqrt{\frac{3 \pm \sqrt{5}}{2}} = 2\cos\frac{\pi}{5},\ 2\cos\frac{2\pi}{5}, \tag{4}$$

$$\beta_\pm = \sqrt{\frac{5 \pm \sqrt{5}}{2}} = 2\cos\frac{\pi}{10},\ 2\cos\frac{3\pi}{10}, \tag{5}$$

so that

$$J_4 = \begin{pmatrix} 1+i\alpha_+ & 0 & 0 & 0 \\ 0 & 1-i\alpha_+ & 0 & 0 \\ 0 & 0 & 1+i\alpha_- & 0 \\ 0 & 0 & 0 & 1-i\alpha_- \end{pmatrix},$$

$$\det J_4 = \left(1+\tfrac{3+\sqrt{5}}{2}\right)\left(1+\tfrac{3-\sqrt{5}}{2}\right) = \tfrac{5+\sqrt{5}}{2} \cdot \tfrac{5-\sqrt{5}}{2} = 5.$$

The 9×9 matrix, with $n+1 = 10$, inherits all of J_4 together with the eigenvalue 1, since 5 divides 10, and adds the four eigenvalues $1 \pm i\beta_{\pm}$, so that, with determinant 55,

$$J_9 = \operatorname{diag}\big(1,\ 1 \pm i\beta_+,\ 1 \pm i\alpha_+,\ 1 \pm i\beta_-,\ 1 \pm i\alpha_-\big).$$

The 8×8 matrix, with $n + 1 = 9$, inherits only $1 \pm i$ from $\mathbb{M}_2$, since 3 divides 9; its other six eigenvalues are $1 \pm 2i\cos\frac{\pi}{9}$, $1 \pm 2i\cos\frac{2\pi}{9}$ and $1 \pm 2i\cos\frac{4\pi}{9}$, where the numbers $2\cos\frac{\pi}{9}$, $-2\cos\frac{2\pi}{9}$ and $-2\cos\frac{4\pi}{9}$ are the three roots of the cubic $x^3 - 3x - 1 = 0$. All three roots are real, yet Cardano's formula can reach them only through cube roots of complex numbers, the *casus irreducibilis*, which is why sympy returns them as `CRootOf` objects or as nested cube roots with imaginary parts that cancel. The 6×6 matrix, with $n+1 = 7$ prime, inherits nothing from a smaller matrix and, being of even order, has no middle eigenvalue 1; its eigenvalues $1 \pm ir$ take r from the three roots $2\cos\frac{\pi}{7}$, $2\cos\frac{3\pi}{7}$, $2\cos\frac{5\pi}{7}$ of $x^3 - x^2 - 2x + 1 = 0$, another irreducible case.

Clarke [5] explores the simplification of square roots of the form $\sqrt{a \pm b\sqrt{m}}$, outlining the connection between Pell's equation and the square roots of surds given the simplification condition:

$$a^2 - mb^2 = n^2, \tag{6}$$

where $a, b,$ and m are integers, m is square-free, and n is an integer. If this condition holds, the surd $\sqrt{a \pm b\sqrt{m}}$ can be expressed as:

$$\sqrt{a \pm b\sqrt{m}} = \frac{\sqrt{a+n} \pm \sqrt{a-n}}{\sqrt{2}},$$

as squaring the right-hand side, $\frac{(a+n)+(a-n)\pm 2\sqrt{a^2-n^2}}{2} = a \pm \sqrt{mb^2}$, confirms. We apply this framework to examine whether simplifications exist for (5).

Case: $\sqrt{3 \pm \sqrt{5}}$: $a = 3$, $b = 1$, $m = 5$ so $a^2 - mb^2 = 3^2 - 5 \cdot 1^2 = 9 - 5 = 4 = 2^2$, so since $n = 2$:

$$\sqrt{3 \pm \sqrt{5}} = \frac{\sqrt{3+2} \pm \sqrt{3-2}}{\sqrt{2}} = \frac{\sqrt{5} \pm 1}{\sqrt{2}}, \qquad \alpha_{\pm} = \frac{1}{\sqrt{2}}\sqrt{3 \pm \sqrt{5}} = \frac{\sqrt{5} \pm 1}{2},$$

which are the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$ and $-\phi' = \frac{\sqrt{5}-1}{2} = \phi^{-1}$, where $\phi' = \frac{1-\sqrt{5}}{2}$ is its conjugate. So $\alpha_+ = \phi$, $\alpha_- = \phi^{-1}$, and we have:

$$J_4 = \begin{pmatrix} 1 + i\phi & 0 & 0 & 0 \\ 0 & 1 - i\phi & 0 & 0 \\ 0 & 0 & 1 + i\phi^{-1} & 0 \\ 0 & 0 & 0 & 1 - i\phi^{-1} \end{pmatrix}, \qquad \det J_4 = (1+\phi^2)(1+\phi^{-2}) = 2 + \phi^2 + \phi^{-2} = 5.$$

Case: $\sqrt{5 \pm \sqrt{5}}$: $a = 5$, $b = 1$, $m = 5$, so $a^2 - mb^2 = 25 - 5 = 20$, which is not a perfect square, and $\sqrt{5 \pm \sqrt{5}}$ within $\beta_{\pm}$ cannot be denested in this way. The two cases differ exactly as Clarke's condition predicts: denesting needs $a^2 - mb^2$ to be a perfect square, a generalised Pell condition on a and b.

The code loopCharcteristic1.ipynb delivers the eigenvalues of the characteristic equation of the complex Fibonacci matrix, collected in Table 2, where n is the order of the matrix. Geometrically recall that these eigenvalues, λ represent the scalar factors by which the matrix (merely) scales its

n	$n+1$	**Eigenvalues**, $\lambda = 1 + 2i\cos\frac{\pi k}{n+1}$	$\mathbf{Fib}_{n+1}$
2	3	$1 \pm i$	2
3	4	$1,\ 1 \pm \sqrt{2}i$	3
4	5	$1 \pm i\phi,\ 1 \pm i\phi^{-1}$	5
5	6	$1,\ 1 \pm i,\ 1 \pm \sqrt{3}i$	8
6	7	$1 \pm ir$, r a root of $x^3 - x^2 - 2x + 1$	13
7	8	$1,\ 1 \pm \sqrt{2}i,\ 1 \pm i\sqrt{2 \pm \sqrt{2}}$	21
8	9	$1 \pm i,\ 1 \pm ir$, r a root of $x^3 - 3x - 1$	34
9	10	$1,\ 1 \pm i\phi,\ 1 \pm i\phi^{-1},\ 1 \pm \frac{i}{2}\sqrt{10 \pm 2\sqrt{5}}$	55
10	11	$1,\ 1 \pm ir$, r a root of $x^5 - x^4 - 4x^3 + 3x^2 + 3x - 1$	89
11	12	$1,\ 1 \pm i,\ 1 \pm \sqrt{2}i,\ 1 \pm \sqrt{3}i,\ 1 \pm \frac{i}{2}(\sqrt{6} \pm \sqrt{2})$	144

Table 2: Eigenvalues and Fibonacci Numbers. Signs $\pm$ inside a radical are independent of the sign in front of i.

eigenvectors under transformation. Each eigenvalue corresponds to a stretching (or compressing) factor in the direction of its eigenvector. For the original matrix $\mathbb{M}$, the action of the matrix on an eigenvector v satisfies $\mathbb{M}v = \lambda v$, where λ is the eigenvalue.
The determinant of J_n as the product of eigenvalues represents the scaled n-dimensional volume element of the space transformed by the Fibonacci matrix, and the inherited eigenvalues show up as inherited factors:

$$n = 2 : \prod_{\lambda_2} = (1 - i)(1 + i) = 2, \tag{7}$$

$$n = 3 : \prod_{\lambda_3} = 1 \cdot (1 - \sqrt{2}i)(1 + \sqrt{2}i) = 3, \tag{8}$$

$$n = 4 : \prod_{\lambda_4} = (1 - i\phi)\,(1 + i\phi)\left(1 - i\phi^{-1}\right)\left(1 + i\phi^{-1}\right) = 5,$$

$$n = 5 : \prod_{\lambda_5} = 1 \cdot (1 - i)\,(1 + i)\,2\Phi' \cdot 2\Phi = 4\prod_{\lambda_2} \Phi' \cdot \Phi = 8, \tag{9}$$

$$\begin{aligned} n = 7 : \prod_{\lambda_7} &= 1 \cdot (1 - \sqrt{2}i)(1 + \sqrt{2}i) \\ &\quad \cdot (1 - i\sqrt{2 - \sqrt{2}})(1 + i\sqrt{2 - \sqrt{2}})\,(1 - i\sqrt{2 + \sqrt{2}})(1 + i\sqrt{2 + \sqrt{2}}) \\ &= \prod_{\lambda_3} (3 - \sqrt{2})(3 + \sqrt{2}) = 3 \cdot 7 = 21, \end{aligned} \tag{10}$$

$$n = 11 : \prod_{\lambda_{11}} = \prod_{\lambda_5} \cdot \prod_{\lambda_3} \cdot (3 + \sqrt{3})(3 - \sqrt{3}) = 8 \cdot 3 \cdot 6 = 144. \tag{11}$$

Here Φ and Φ' are the complex golden ratios, the roots of the quadratic equation $x^2 - x + 1 = 0$,

$$\Phi = \frac{1 + i\sqrt{3}}{2}, \quad \Phi' = \frac{1 - i\sqrt{3}}{2}, \tag{12}$$

so that $1 \pm \sqrt{3}\,i = 2\Phi,\ 2\Phi'$ and $\Phi\Phi' = 1$. In the last line the factor $6 = (3 + \sqrt{3})(3 - \sqrt{3})$

comes from the two pairs of eigenvalues $1 \pm \frac{i}{2}(\sqrt{6} \pm \sqrt{2})$ that are new at $n+1 = 12$, since $1 + \frac{1}{4}(\sqrt{6} \pm \sqrt{2})^2 = 3 \pm \sqrt{3}$.
These products can be viewed as $\mathbb{C}$omplex **field extension** decompositions: an integer such as $\mathcal{F}_{n+1}$, irreducible as a whole number, factors once the eigenvalues are admitted. The determinant thus intertwines Fibonacci numbers with polynomial characteristics through its action on the Fibonacci-Jordan matrix. This is to be expected as the breaking down of polynomials into irreducible factors parallels decomposing composite numbers into prime factors.

Accordingly we gather some observations of the Galois theory of field extensions:

1. **Decomposition of Composite Numbers and Polynomials:**
   Polynomials $f(x) \in F[x]$ over a **field** F can be expressed as a product of irreducible polynomials, as integers can be expressed as a product of primes. While composite numbers n decompose uniquely into prime factors:

   $$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}.$$

   composite polynomials $f(x)$ decompose uniquely (up to units) into irreducible factors:

   $$f(x) = p_1(x)^{e_1} p_2(x)^{e_2} \cdots p_k(x)^{e_k}.$$

2. **Factorization and Rings:** analogous to the **fundamental theorem of arithmetic**, the **unique factorization theorem** ensures this decomposition in $F[x]$. For integers, factorization occurs in the **ring** $\mathbb{Z}$, where primes are indivisible except by ± 1 and themselves. Extending to the **Gaussian integers** $\mathbb{Z}[i]$, which include complex numbers of the form $a + bi$ (where $a, b \in \mathbb{Z}$ and $i = \sqrt{-1}$), the notion of primes changes.

3. **Root extensions and splits:** integers $\mathbb{Z}$-primes either split or remain inert within **field extensions**, depending on the properties of the extension. In $\mathbb{Z}[i]$, $\mathbb{Z}$-primes are redefined to account for the new units $\pm i$, so that factorization by **Gaussian primes** means 2, 5 lose their primality status:

   $$2 = (1+i)(1-i), \quad 5 = (2+i)(2-i).$$

   Primes in $\mathbb{Z}[i]$ are thus classified into:

   (a) **Real primes** that remain irreducible (e.g., 3 retains its primality, remaining inert in $\mathbb{Z}[i]$ as it cannot be decomposed into Gaussian prime factors).

   (b) **Decomposable primes**, such as 2 and 5, which split into products of Gaussian primes.

   Extending further to $\mathbb{Q}(i, \sqrt{2})$, we lose the primality of 3. The eigenvalues $[1, 1-\sqrt{2}i, 1+\sqrt{2}i]$ for $n = 3$ involve $\sqrt{2}$ and i so $3 = (1-\sqrt{2}i)(1+\sqrt{2}i)$ is factorizable in $\mathbb{Q}(i, \sqrt{2})$ (in accordance with the decomposition of its characteristic polynomial into linear terms).

   For polynomials, factorization occurs in $F[\lambda]$, but the irreducibility depends on the field F. For instance, $\lambda^2 - 2$ is irreducible over $\mathbb{Q}$, the field of rational numbers but factors as $(\lambda - \sqrt{2})(\lambda + \sqrt{2})$ over $\mathbb{R}$:

   $$\lambda^2 - 2 \quad \text{(irreducible over } \mathbb{Q}\text{) splits as} \quad (\lambda - \sqrt{2})(\lambda + \sqrt{2}) \quad \text{over } \mathbb{R}.$$

4. **Galois Theory:** formalizes these ideas by studying the symmetry of roots:

$$M \sim \text{Primes} \quad \text{and} \quad \text{Galois Group} \sim \text{Field Extensions}.$$

A composite number's prime factorization reflects its divisors and the structure of its multiplicative group. Similarly, the splitting field of a polynomial reflects how its roots relate under symmetries. The **Galois group** of a polynomial captures the structure of the field extension required for its roots, akin to how composite numbers decompose into primes. For the Fibonacci matrices the relevant fields are generated by i and $\cos\big(\pi/(n+1)\big)$, and since $\mathbb{Q}\big(\cos(\pi/N)\big)$ sits inside the field of $2N$th roots of unity, its Galois group is abelian. Every eigenvalue in Table 2 can therefore be written in radicals, even the ones that come from a quintic at $n = 10$, although the radicals may be too unwieldy for `simplify` to find.

Eisenstein Integers

The Eisenstein integers, denoted $\mathbb{Z}[\omega]$, form a quadratic integer ring defined as:

$$\mathbb{Z}[\omega] = \{r + s\omega : r, s \in \mathbb{Z}\},$$

where $\omega = e^{2\pi i/3}$ is a primitive third root of unity. The root ω satisfies:

$$\omega^3 = 1, \quad \omega^2 + \omega + 1 = 0, \quad \omega^2 = -1 - \omega.$$

The Eisenstein integers are a natural generalization of the Gaussian integers $\mathbb{Z}[i]$, which are formed by adjoining i (a fourth root of unity) to $\mathbb{Z}$. Similarly, the Eisenstein integers are formed by adjoining ω (a third root of unity) to $\mathbb{Z}$. The cube roots of unity $1, \omega, \omega^2$ divide the unit circle into three equal arcs and form an equilateral triangle, and ω, ω^2 are the roots of $x^2 + x + 1 = 0$. The complex golden ratios $\Phi = -\omega^2 = 1 + \omega$ and $\Phi' = -\omega = 1 + \omega^2$ are the roots of $x^2 - x + 1 = 0$, the primitive sixth roots of unity, and they are themselves Eisenstein integers, indeed units, so no division by 2 is needed to accommodate $\Phi = \frac{1+i\sqrt{3}}{2}$. Closure under division requires the field of fractions

$$\mathbb{Q}(\omega) = \{r + s\omega : r, s \in \mathbb{Q}\} = \mathbb{Q}(\sqrt{-3}),$$

which contains $i\sqrt{3} = 1 + 2\omega$ but not $\sqrt{3}$ itself. The Eisenstein integers inherit many properties from the Gaussian integers, with modifications due to the third root of unity ω. For $z = r + s\omega$, where $r, s \in \mathbb{Z}$, the following operations are defined:

- **Addition:** Simply add the components:

$$z_1 + z_2 = (r_1 + r_2) + (s_1 + s_2)\omega.$$

- **Multiplication:** Incorporates the property $\omega^2 = -1 - \omega$:

$$z_1 \cdot z_2 = (r_1 r_2 + s_1 s_2(-1 - \omega)) + (r_1 s_2 + r_2 s_1)\omega.$$

- **Norm:** The norm of $z = r + s\omega$ is defined as:

$$N(z) = z \cdot \overline{z} = r^2 - rs + s^2,$$

where $\overline{z} = r - s - s\omega$ is the conjugate of z.

The Eisenstein integers $\mathbb{Z}[\omega]$ generalize the Gaussian integers by introducing the third root of unity ω and its associated algebraic structure. The norm $N(z) = r^2 - rs + s^2$ plays a central role in understanding the structure and arithmetic of Eisenstein integers. It is always a non-negative integer, providing a measure of the "size" of z in $\mathbb{Z}[\omega]$, and it is multiplicative, $N(z_1 z_2) = N(z_1)N(z_2)$. Division with a remainder of smaller norm is always possible, so $\mathbb{Z}[\omega]$ is a Euclidean domain and hence a unique factorization domain (UFD), just like $\mathbb{Z}[i]$. The prime 3 ramifies, $3 = -\omega^2(1-\omega)^2$, and the Eisenstein prime $1-\omega$, of norm 3, appears in the next section.

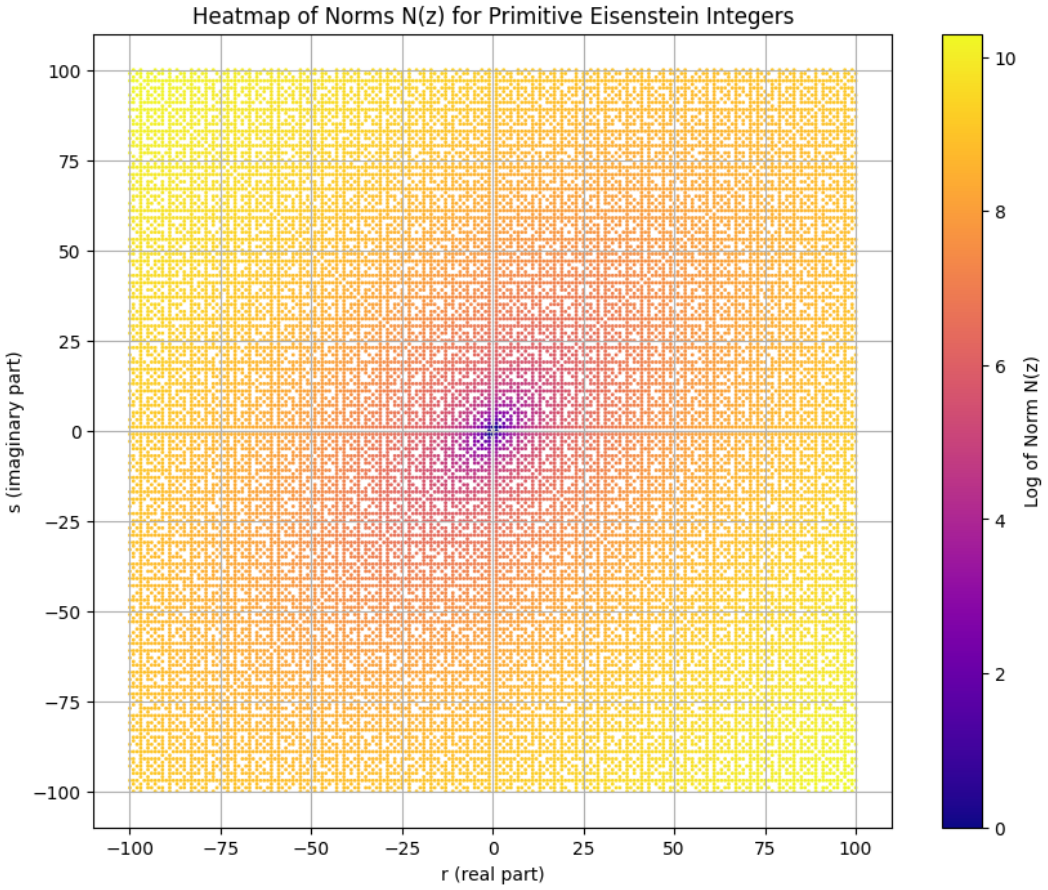


Figure 1: Pythagorean Triples from the elevenths.

The plot visualizes the distribution of **primitive Eisenstein integers** $z = r + s\omega$, where ω is a primitive third root of unity ($\omega^3 = 1$). The key elements of the plot are:

- Each point (r, s) represents the real part r and imaginary coefficient s of Eisenstein integer z.
- The Eisenstein integers shown are **primitive**, meaning they satisfy the condition:
$$\gcd(a, b, c) = 1,$$
where $a = r$, $b = -r + s$, and $c = r^2 - rs + s^2$ (the norm of z).
- The **color intensity** of each point corresponds to the logarithm of the norm:
$$N(z) = r^2 - rs + s^2.$$
This scaling highlights variations in the norm across a wide range of values.

Characteristic Polynomial of $\mathbb{M}_2(1)$ with ω

The matrix $\mathbb{M}_2(1)$ is given by:

$$\mathbb{M}_2(1) = \begin{pmatrix} 1 & \omega \\ \omega & 1 \end{pmatrix}.$$

The characteristic polynomial is computed as:

$$\det(\mathbb{M}_2(1) - \lambda I) = \det \begin{pmatrix} 1-\lambda & \omega \\ \omega & 1-\lambda \end{pmatrix}.$$

Expanding the determinant, we have:

$$\det(\mathbb{M}_2(1) - \lambda I) = (1-\lambda)^2 - \omega^2.$$

Using $\omega^2 = -1 - \omega$ (from $\omega^2 + \omega + 1 = 0$), substitute ω^2 into the determinant:

$$(1-\lambda)^2 - (-1-\omega) = (1-\lambda)^2 + 1 + \omega.$$

Simplify:

$$(1-\lambda)^2 + 1 + \omega = \lambda^2 - 2\lambda + 2 + \omega.$$

Thus, the characteristic polynomial is:

$$\lambda^2 - 2\lambda + 2 + \omega = 0.$$

The eigenvalues are given by solving:

$$\lambda = \frac{2 \pm \sqrt{z}}{2}, \quad \text{where } z = 4 - 4(2+\omega) = -4 - 4\omega = -2 - 2i\sqrt{3}.$$

Factor out -2 from z:

$$z = -2(1 + i\sqrt{3}).$$

Express $1 + i\sqrt{3}$ in polar form:

$$1 + i\sqrt{3} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 2e^{i\pi/3}.$$

Thus:

$$z = -2 \cdot 2e^{i\pi/3} = -4e^{i\pi/3}.$$

Using the polar form, a square root of z is:

$$\sqrt{z} = \sqrt{-4} \cdot \sqrt{e^{i\pi/3}} = 2i \cdot e^{i\pi/6} = 2i\left(\frac{\sqrt{3}}{2} + i\frac{1}{2}\right) = i\sqrt{3} - 1.$$

Substitute $\sqrt{z} = i\sqrt{3} - 1$ into the eigenvalue formula:

$$\lambda = \frac{2 \pm (i\sqrt{3} - 1)}{2}.$$

Compute the two eigenvalues:

$$\lambda_1 = \frac{2 + (i\sqrt{3} - 1)}{2} = \frac{1 + i\sqrt{3}}{2}, \qquad \lambda_2 = \frac{2 - (i\sqrt{3} - 1)}{2} = \frac{3 - i\sqrt{3}}{2}.$$

In Eisenstein terms these are $\lambda_{1,2} = 1 \pm \omega$, which could have been read off at once, because $\mathbb{M}_2(1) = I + \omega P_2$ and P_2 has eigenvalues ± 1 on the eigenvectors $(1,1)$ and $(1,-1)$, exactly as for $\mathbb{M}(2)_1$. The first eigenvalue is the complex golden ratio $\lambda_1 = 1 + \omega = \Phi$, a unit, while the second, $\lambda_2 = 1 - \omega = \sqrt{3}\, e^{-i\pi/6}$, is not Φ' but the Eisenstein prime of norm $N(1-\omega) = 1 + 1 + 1 = 3$. Their product is the determinant, $\Phi(1-\omega) = 1 - \omega^2 = 2 + \omega$, of norm $4 - 2 + 1 = 3$. The matrix $\mathbb{M}_2(1)$ diagonalizes as:

$$\mathbb{M}_2(1) = P \begin{pmatrix} \Phi & 0 \\ 0 & 1-\omega \end{pmatrix} P^{-1}, \qquad P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

so that replacing i by ω as the coupling turns the Gaussian factorisation $2 = (1+i)(1-i)$ of $\det \mathbb{M}(2)_1$ into a unit times the Eisenstein prime above 3.

Our nuanced interplay between primes over field extensions and the roots of characteristic polynomials can be summarized as:

1. **Field Extensions for Eigenvalues** of the characteristic polynomials for each n of Fibonacci matrix lie in specific algebraic nested extensions of $\mathbb{Q}$:
   - For $n = 2$, the eigenvalues $1 - i$ and $1 + i$ lie in $\mathbb{Q}(i)$, the smallest extension of $\mathbb{Q}$ containing i, the imaginary unit.
   - For $n = 3$, the eigenvalues $1 - \sqrt{2}i$ and $1 + \sqrt{2}i$ lie in $\mathbb{Q}(i, \sqrt{2})$, an extension formed by adjoining i and $\sqrt{2}$ to $\mathbb{Q}$. Primes $p \equiv 3 \pmod 4$, such as $3, 7, 11, 19, \ldots$, are irreducible in $\mathbb{Z}[i]$, and 3 and 11 split once $\sqrt{-2}$ is adjoined, as $3 = (1 - \sqrt{2}i)(1 + \sqrt{2}i)$ and $11 = (3 - \sqrt{2}i)(3 + \sqrt{2}i)$.
   - For $n = 7$, eigenvalues such as $1 \pm i\sqrt{2 - \sqrt{2}}$ require the nested radical and the imaginary unit, in the field extension $\mathbb{Q}(i, \sqrt{2 - \sqrt{2}})$, which already contains $\sqrt{2} = \sqrt{2 + \sqrt{2}}\,\sqrt{2 - \sqrt{2}}$ and $\sqrt{2 + \sqrt{2}}$.
2. **Inclusion of Smaller Roots:** From Table 2, we observe a hierarchy of eigenvalues in which the roots of a smaller matrix J_d appear within those of a larger J_n exactly when $d+1$ divides $n+1$:
   - The eigenvalues of J_2 $(1 \pm i)$ are present in J_5, J_8 and J_{11}, since 3 divides 6, 9 and 12, but not in J_7.
   - The eigenvalues of J_3 $(1 \pm \sqrt{2}i)$ appear in J_7 and J_{11}, since 4 divides 8 and 12.
   - The eigenvalues of J_4 appear in J_9, since 5 divides 10, and all of J_5 appears in J_{11}, since 6 divides 12.

   This nesting of roots reflects the structure of field extensions, with each J_n encapsulating the extensions required by the smaller Fibonacci matrices whose $d+1$ divides its $n+1$, and it is the eigenvalue form of the divisibility $\mathcal{F}_{d+1} \mid \mathcal{F}_{n+1}$.

The decomposition into complex roots of the eigenvalues of Fibonacci matrices reveals the hierarchical nature of field extensions and provides a rich framework for pedagogical exploration. The Galois groups of these polynomials encode the symmetries of the roots, and the polynomials of Table 1 could be further analyzed for their splitting fields, irreducibility, and symmetry properties. The orders whose eigenvalues look structurally simple are those for which $n+1$ has many small

divisors, such as $n = 5, 7, 11$, while the orders that defeat python's sympy `simplify` algorithm within generalEigenVectorFibonacci.ipynb are those for which $n + 1$ is prime, or a power of an odd prime, such as $n = 6, 8, 10$, where $\cos\big(\pi/(n+1)\big)$ satisfies an irreducible cubic or quintic. Writing every eigenvalue as $1 + i(\zeta + \zeta^{-1})$ with ζ a $2(n+1)$th root of unity makes the roots of unity explicit, and is the natural first step in the further exploration.

Revision note (October 2026). This version corrects the following in the original: the eigenvalues of J_4, previously given as $1 \pm \frac{i}{\sqrt{2}}\phi$ because of a missing $\sqrt{2}$ in Clarke's denesting formula; the matrix previously labelled J_8, whose entries were eigenvalues of J_9; the lower coefficients of the characteristic polynomials for $n \geq 5$ in Table 1; the eigenvalue entries for $n = 6, 8, 10$ in Table 2, which lay outside the band $|\operatorname{Im}\lambda| < 2$; the metallic matrices, which need m on the diagonal and i beside it; the naming of the sequence $1, 1, 3, 7, 17, 41$; the second eigenvalue of the ω-matrix; the description of $\mathbb{Q}(\omega)$ and of Φ and Φ'; and the claim that simplicity follows primality of n, which is replaced by the role of $n + 1$.

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Appendix: Binet's formula from M_1

Following the exposition of Rochford, [4] we have

$$\text{for } M_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \text{ so that } \begin{pmatrix} F_2 \\ F_1 \end{pmatrix} = M_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

and for $n \geq 1$,

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}, \quad \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = M_1^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad F_n = \begin{pmatrix} 0 & 1 \end{pmatrix} M_1^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$$\begin{pmatrix} F_3 \\ F_2 \end{pmatrix} = M_1 \begin{pmatrix} F_2 \\ F_1 \end{pmatrix} = M_1^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} F_4 \\ F_3 \end{pmatrix} = M_1 \begin{pmatrix} F_3 \\ F_2 \end{pmatrix} = M_1^3 \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

To diagonalize M_1 is to write it as $M_1 = PDP^{-1}$, where D is a diagonal matrix:

$$M_1^2 = PDP^{-1}PDP^{-1} = PD^2P^{-1}.$$

Similarly, for any natural number n, we have $M_1^n = PD^nP^{-1}$. We first calculate its eigenvalues by finding the roots of the characteristic polynomial, $\det(M_1 - \lambda I)$, as

$$\lambda_1 = \frac{1+\sqrt{5}}{2} \quad \text{and} \quad \lambda_2 = \frac{1-\sqrt{5}}{2}.$$

The matrix D is therefore

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

We note that $\lambda_1 + \lambda_2 = 1$ and $\lambda_1\lambda_2 = -1$. The columns of P are the eigenvectors corresponding to the eigenvalues that form the diagonal entries of D and so we calculate a basis for the null space of $M_1 - \lambda I$. For the eigenvalue λ_1, using $1 - \lambda_1 = \lambda_2$ and $\lambda_1\lambda_2 = -1$,

$$M_1 - \lambda_1 I = \begin{pmatrix} \lambda_2 & 1 \\ 1 & -\lambda_1 \end{pmatrix} \sim \begin{pmatrix} 1 & -\lambda_1 \\ 0 & 0 \end{pmatrix},$$

where the second row has been cleared by subtracting λ_2 times the reordered first row, since $1 + \lambda_1\lambda_2 = 0$. The null space of $M_1 - \lambda_1 I$ is spanned by the vector $\begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix}$, and likewise that of $M_1 - \lambda_2 I$ by $\begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix}$. Taking these as the columns of P, its determinant is $\lambda_1 - \lambda_2 = \sqrt{5}$, and

$$P = \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix}, \quad P^{-1} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{pmatrix}.$$

(The two eigenvectors are orthogonal, since $\lambda_1\lambda_2 + 1 = 0$, so one could instead normalise each to unit length and use $P^{-1} = P^\top$; the lengths are $\sqrt{\lambda_1^2 + 1}$ and $\sqrt{\lambda_2^2 + 1}$, which differ, so the factor is not a common $1/\sqrt{5}$.) Finally, $P^{-1}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, and the bottom row of P is $(1 \;\; 1)$, so we derive the closed-form expression for F_n:

$$F_n = \begin{pmatrix} 0 & 1 \end{pmatrix} PD^nP^{-1}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{5}}\left(\lambda_1^n - \lambda_2^n\right).$$

This completes the derivation of Binet's formula for the Fibonacci sequence:

$$F_n = \frac{1}{\sqrt{5}}\left(\lambda_1^n - \lambda_2^n\right).$$